

# Sample End Semester Question Paper - 2025

**Subject: Engineering Mathematics I | Code: MA101 | Semester: 1**

**Branch: CSE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks:**

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).  
It is not an official university paper and is provided to demonstrate the platform.

## **PART A - Short questions (10 x 1 = 10 Marks)**

1. Define the gamma function.
2. State one property of the beta function.
3. What are the Dirichlet conditions?
4. State the sufficient condition for maxima and minima of a function of two variables.
5. What is a homogeneous differential equation?
6. State Euler's theorem for homogeneous functions.
7. What is the Jacobian?
8. State Taylor's theorem for one variable.
9. What is an exact differential equation?
10. State Fourier's theorem briefly.

## **PART B - Long questions (10 x 8 = 80 Marks)**

1. Explain total derivative and implicit differentiation with examples.
2. Solve a second order linear ODE with constant coefficients.
3. Explain partial derivatives and prove Euler's theorem with an example.
4. Find the maxima and minima of a given function of two variables.
5. Expand a given function using Taylor's and Maclaurin's series.
6. Solve first order ODEs: separable, homogeneous and exact, with examples.
7. Find the Fourier series of a given periodic function.
8. Solve a Bernoulli equation and a linear first order ODE with examples.
9. Explain gamma and beta functions. Evaluate a given integral using them.
10. Evaluate a double integral and change the order of integration.